



LA GRANDEE INTERNATIONAL COLLEGE

Simalchaur Pokhara, Nepal

A Final Report

On

An Inventory Management System

“StockEase”

Submitted to

LA GRANDEE International College

Bachelor of Computer Application (BCA) Program

In partial fulfilment of the requirements for the degree of Bachelor of Computer
Application (BCA) under Pokhara University

Submitted by

Name	Program	Semester	P.U Registration No.
Yubaraj Pandey	BCA	6 th	2021-1-53-0373
Pawan Poudel	BCA	6 th	2021-1-53-0357

Date: 01/03/2025

Acknowledgement

We would like to express our sincere gratitude to the college and the respective individual who have helped us to make this project possible. Without them the project would not have been completed or possible. Their constant guidance and cooperation have made us to achieve the goals which we were pursuing to achieve while selecting the project.

We are grateful to our project supervisor **Mr. Sunil Sapkota** and BCA Coordinator **Mr. Kundan Chaudhary** for their guidance and mentorship throughout the project.

We are also appreciative among each other as we believed in teamwork and achieved the project in the given time.

Sincerely,

Pawan Poudel

Yubaraj Pandey

Student Declaration

We, Pawan Poudel and Yubaraj Pandey, students of 6th Semester hereby declare that we are only the authors of this project proposal and we are submitting this project to LA GRANDEE International College, Faculty of Science and Technology, Pokhara University in the partial fulfilment of the requirement of Bachelor in Computer Application under the supervision of Sir Sunil Sapkota and that no resources other than those listed here have been used.

Student Name	Pawan Poudel	Yubaraj Pandey
Registration No	2021-1-53-0357	2021-1-53-0373
Exam Roll No	21535158	21535172
Date	01/03/2025	01/03/2025
Signature	 	

Supervisor's Declaration

I hereby declare that the project entitled “**Inventory Management System-StockEase**” has been carried out under my direct supervision by Yubaraj Pandey and Pawan Poudel during their sixth semester for the partial fulfilment of the requirements for the degree of BCA program under Pokhara University is completed to my satisfaction and be processed for final evaluation.

Mr. Sunil Sapkota

Date: 01/03/2025

Letter of Approval

We certify that we have examined this report entitled “**Inventory Management System-StockEase**” and are satisfied with the project defence. It is satisfactory in the scope and qualify as project in partial fulfilment of the requirements for the degree of **Bachelor in Computer Application (BCA)** under **Pokhara University**.

Supervisor

Mr. Sunil Sapkota

Examiner

Er. Asmit Nepali

Program Coordinator

Mr. Kundan Chaudhary

Date: 01/03/2025

Abstract

The “**Inventory Management System**” is a web-based application developed using HTML, CSS, JavaScript, PHP, and MySQL to simplify and streamline inventory tracking and management. The system enables users to manage stock levels, add or update product details, monitor inventory movement, and generate reports through a responsive and user-friendly interface

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Abbreviations

API	Application programming interface
Css	Cascading style sheet
DFD	Data flow diagram
E-commerce	Electronic Commerce
E.g	Example
ER	Entity relationship
GCC	GNU Compiler Collection
HTML	Hypertext Markup Language
IDE	Integrated Development Environment
IMS	Inventory Management System
Mysql	My Structured Query Language
PHP	Hypertext Preprocessor
Rfid	Radio- Frequency Identification

1. Introduction

StockEase is an all-in-one solution designed to streamline stock and inventory management for businesses of all sizes. Whether you're managing a small retail store, a large warehouse, or an e-commerce platform, the platform provides real-time insights into stock levels, minimizing the risk of overstocking or running out of key products. Its user-friendly interface allows for smooth navigation through intuitive search filters, enabling precise tracking based on product category, location, quantity, and stock status.

StockEase is a web-based application that allows users to efficiently manage inventory based on their business needs. It also provides suppliers with an easy-to-use platform for uploading and managing their product listings. Users can easily filter inventory details and track stock movements, helping them make informed decisions about replenishing stock or managing surplus. Whether you're overseeing raw materials for manufacturing or managing finished goods in retail, StockEase ensures complete transparency in stock levels, sales performance, and order fulfilment.

The platform's motto, "Optimize Your Inventory, Maximize Your Efficiency," reflects its commitment to reducing manual errors, improving accuracy, and saving time. With real-time data and secure messaging, StockEase connects businesses with suppliers, simplifying the ordering process while allowing users to analyse price trends and negotiate deals. By offering complete transparency and automating repetitive tasks, StockEase saves valuable time for managers and suppliers alike, ensuring smooth operations across the supply chain.

2. Problem Statement

- Inaccurate stock tracking and manual errors disrupt operations.
- Poor integration and limited visibility hinder efficiency.

3. Objectives

In the current digital landscape, businesses are increasingly turning to computerized solutions to enhance efficiency and streamline operations. An "Inventory Management System" is designed to meet this demand, providing various features that help improve business performance and optimize daily operations. Built to address key challenges, the main objective of this project is:

- **Enhanced Inventory Control:** Ensure accurate tracking of stock levels, product availability, and sales transactions to prevent stock shortages and overstocking, improving operational efficiency.
- **Data-Driven Financial Monitoring:** Implement a robust financial tracking system that records sales transactions and generates detailed sales reports, enabling accurate financial analysis and informed business decisions

4. Background Study

Inventory management has long been a critical function in businesses, playing a central role in ensuring that the products are available at the right time and in the right quantity. As technology has advanced, inventory systems have evolved from manual processes to highly sophisticated solutions. IMS Software by IMS Group in Nepal exemplifies this evolution by offering comprehensive tools tailored for businesses in diverse industries, including retail, hospitality, and manufacturing.

The evolution of technology introduced computerized inventory management systems, which offered significant improvements over manual methods. Early systems were primarily used by large-scale businesses to track stock levels, automate reordering processes, and reduce excess inventory. Over time, the adoption of inventory management systems expanded across industries, from retail and manufacturing to healthcare and hospitality, as more businesses recognized the need for precise inventory control to stay competitive.

Modern inventory management systems offer a wide range of features, such as real-time stock tracking, integration with sales and procurement systems, barcode and RFID scanning, and predictive analytics. These systems enable businesses to maintain optimal stock levels, reduce waste, and improve order fulfillment processes. Cloud-based solutions and mobile applications have further enhanced accessibility and ease of use, allowing businesses to manage their inventory from anywhere with real-time updates.

As businesses continue to face challenges such as fluctuating demand, supply chain disruptions, and changing customer expectations, the role of a robust inventory management system becomes even more crucial. An effective system helps businesses minimize stockouts, optimize warehouse operations, and provide accurate data for better decision-making, ultimately leading to improved customer satisfaction and increased profitability.

5. Requirement document

The purpose of this document is to outline the functional and non-functional requirements for an Inventory Management System. This document will be used as a guide for the development team during the design and implementation phases of the project.

5.1 Functional Requirements:

Registration and Login:

- New users can register and log in using secure credentials.

Inventory Management:

- Add, edit, delete, and view inventory items.
- Update stock levels automatically when items are sold or received.

Product Categorization:

- Categorize products based on type, brand, or other attributes.

Search and Filters:

- Allow users to search for products or filter items by name, category etc.

Reporting and Analytics:

- Generate reports for inventory status.

5.2 Non-Functional requirements

Non-Functional requirements are set of specifications that describe the system's operation capabilities and constraints and attempt to improve its functionality.

Some non-functional requirements are:

Security:

- User passwords must be stored securely using encryption.

- The system must ensure secure payment processing by integrating with a reliable payment gateway.
- Implement user authentication to prevent unauthorized access.

Usability:

- The interface should be intuitive and easy to use for customers with basic computer or smartphone skills.
- The system should provide clear error messages and guidance for users when issues occur (e.g., invalid login credentials or unavailable seats).

Maintainability:

- The system should be modular, making it easy to update or add new features

These non-functional requirements ensure that the system meets the expectations of performance, reliability, security, and usability, contributing to a smooth experience for users.

5.3 Technical Requirement

Technical Requirements

Technical requirements are necessary technical capabilities and resources needed for a system to be made and function properly.

R.ID	Requirement	Description of Requirement Statement	Priority (Must, Want)	Comments
1	Programming Language	PHP, JavaScript, HTML, CSS	Must	Backend in PHP, Frontend in JS, HTML, CSS
2	Operating System	Cross-platform (Windows, Linux, macOS)	Must	
3	Database	MySQL or SQLite	Must	
4	User Interface	Web-based	Must	
5	Hosting	Apache/Nginx Server	Want	
6	Storage	10GB	Must	
7	Security	GCC	Want	
8	Development Environment (IDE)	VS Code, XAMPP	Want	Recommended for development
9	Security	User authentication & data encryption	Must	Secure login and user data protection
10	Scalability	Supports increasing users and customers	Must	Should handle future expansion

Table 1: Technical Requirement

6. System Design

Dataflow, Entity-relationship diagram are used for understanding the system's design and its functionalities, and both are important for creating proper documentation.

6.1 Data Flow Diagram

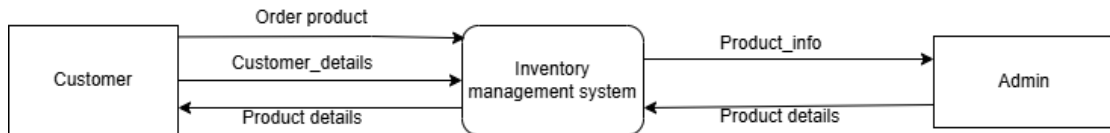


Fig 6. 1: Level 0 DFD

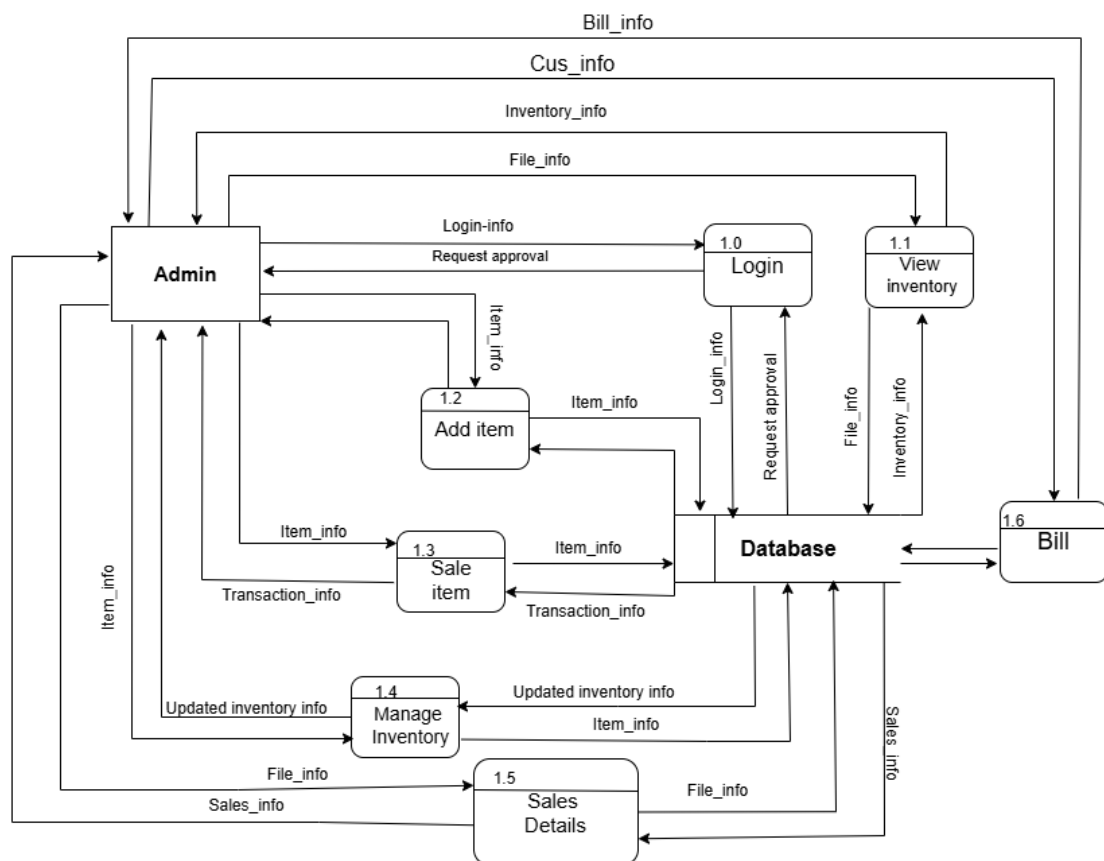


Fig 6. 2: Level 1 DFD

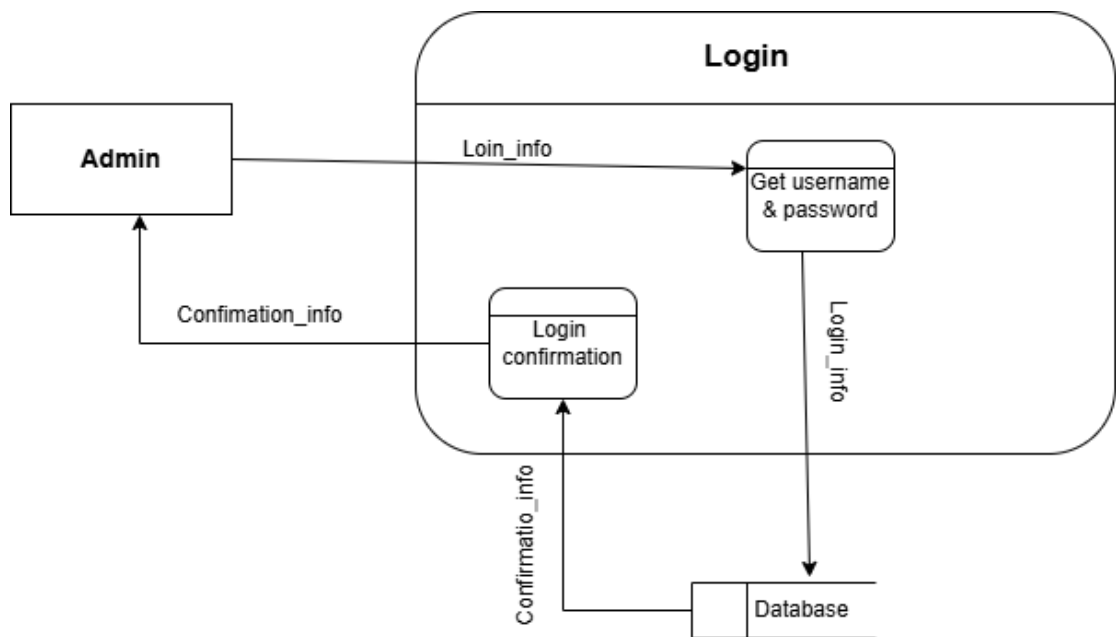


Fig 6. 3: Level 2 DFD for Login

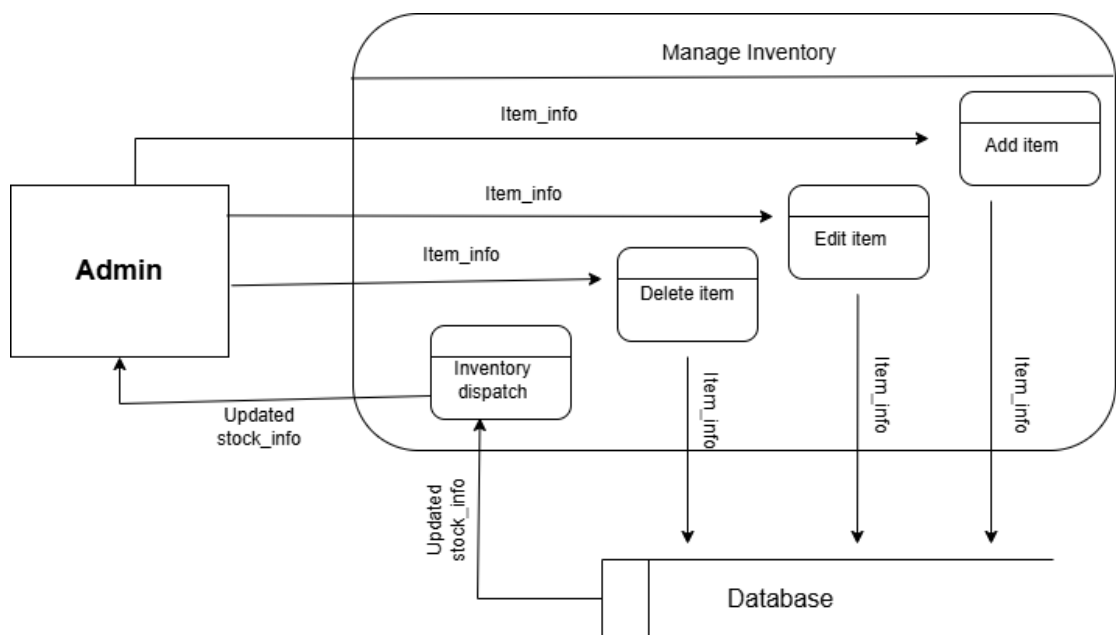


Fig 6. 4: Level 2 DFD for Inventory

6.2 Entity Relationship Diagram (ER-Diagram)

An Entity Relationship Diagram is a diagram that represents relationships among entities in a database. It is commonly known as an ER Diagram.

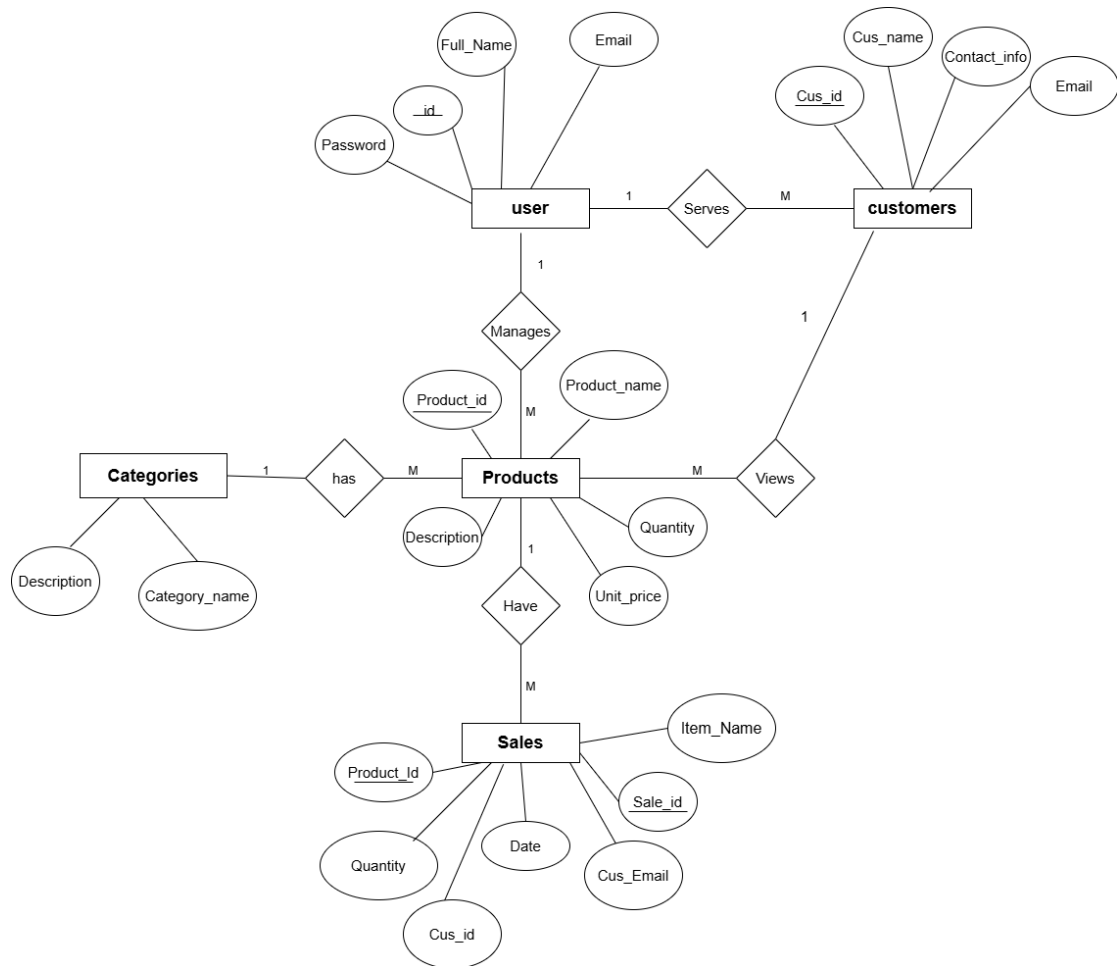


Fig 6. 5: ER Diagram

7. Development

Project development is the process of planning, designing and executing a specific task to achieve a defined outcome or goal. Development provides guidance to develop and build a project in specific time. Software Development includes different phases like feasibility analysis, Requirement Analysis, System design, coding and Module Testing, Integration and Delivery deployment with Documentation

7.1 Development Methodology

We are employing the **Agile Methodology** as our chosen approach for project development. This model follows an iterative process that emphasizes flexibility, collaboration, and continuous improvement. Agile is divided into short development cycles called *sprints*, which typically last 1-4 weeks. Each sprint involves planning, development, testing, and review, allowing teams to deliver incremental updates to the project.

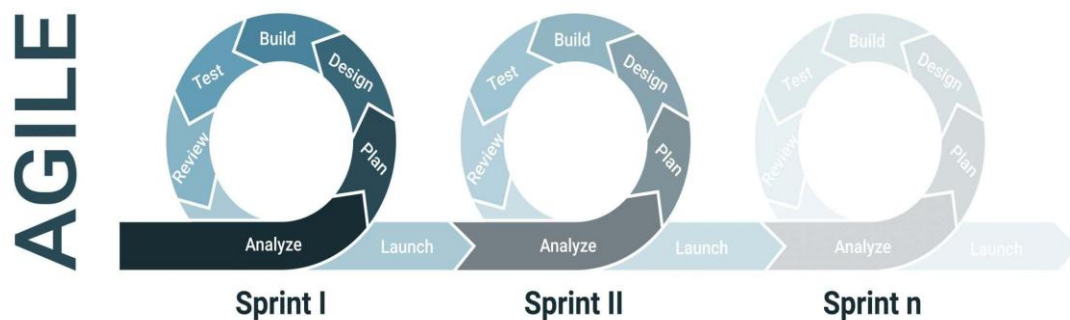


Fig 7. 1: Agile Methodology

The project was divided into five sprints, each sprint focuses on specific features.

Sprint 1: User Registration and Authentication

- **Frontend:** Developed registration and login forms using HTML/CSS, with JavaScript for client-side validation.
- **Backend:** Implemented PHP scripts for user authentication, including session management.
- **Database:** Created a User's table with encrypted passwords for secure storage.
- **Testing:** Verified successful user login/logout and session handling.

Sprint 2: Inventory Management Dashboard

- **Frontend:** Designed a responsive admin dashboard to display stock levels, item details, and categories.
- **Backend:** Used PHP to fetch and display inventory data from the Inventory table.
- **Database:** Created tables for storing product information, stock levels, and categories.
- **Testing:** Ensured real-time updates of inventory data on the dashboard.

Sprint 3: Stock Update and Tracking

- **Frontend:** Implemented forms for adding, updating, and removing stock items.
- **Backend:** Developed PHP scripts to handle stock transactions (e.g., adding new stock, editing existing stock).
- **Database:** Ensured stock updates reflect accurately in the Inventory table.
- **Testing:** Verified that stock changes are recorded correctly and prevent overstocking/understocking.

Sprint 4: Order Processing and Billing

- **Frontend:** Created a checkout process for employees to handle customer orders.
- **Backend:** Used PHP to process orders, generate invoices, and deduct stock from the inventory.
- **Database:** Linked orders with stock levels to update inventory upon purchase.
- **Testing:** Ensured correct order calculations and stock updates after purchases.

Sprint 5: Reporting and Analytics

- **Frontend:** Added a dashboard section for generating sales and inventory reports.
- **Backend:** Developed PHP scripts to fetch and display report data dynamically.
- **Database:** Created tables for storing order history and transaction records.

- **Testing:** Validated report accuracy for insights into stock movement and sales trends.

7.2 Project Gantt Chart

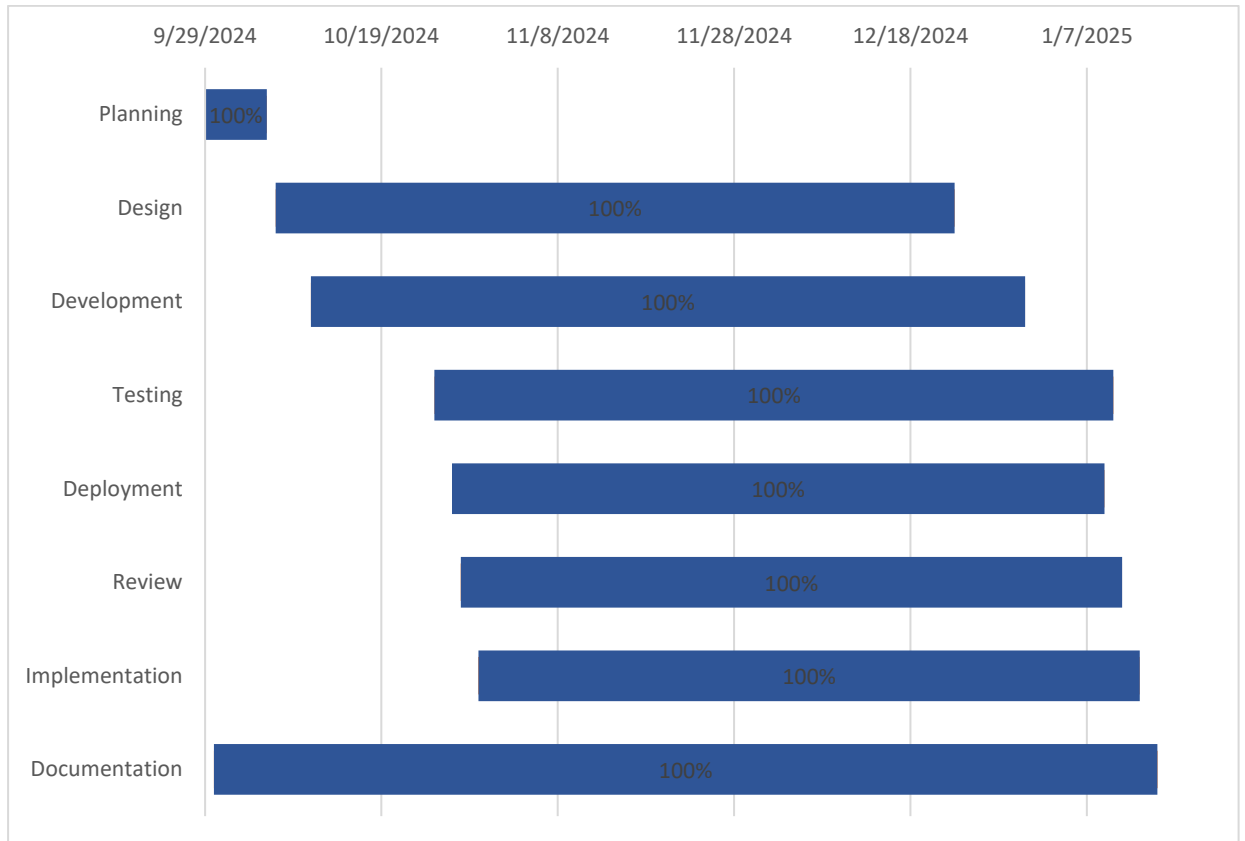


Fig 7. 2: Project Gantt Chart

8. Testing

Software testing is the process of evaluating a software product to ensure it functions as intended. It involves running test cases to identify bugs and verify that the software meets expectations.

We had performed following test cases:

Test Cases

Test Case 1

Title: Module testing of a Login function– Authentication of users

Description: A user should be able to log in with a registered email and password.

Precondition: The user and admin must have an email and password that is registered.

Test Case#	Test Case	Test Steps	Test Data	Expected Results
001.	Verify Login	1. Go to Login 2. Enter Email 3. Enter Password 4. Click Login	email = ab Password= my	Login Unsuccessful
002.	Verify Login	1. Go to Login 2. Enter email 3. Enter Password 4. Click Login	email = aa Password= aa	Login Successful

003.	Verify Login	1. Go to Login 2. Enter email 3. Enter Password 4. Click Login Repeat the process for 3 times	email =aa Password=123	Login Unsuccessful.
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Table 2: Test case for login

Test Case 2:

Title: Module Testing of Dashboard function –Add Products

Description: An admin should be able to add the new products.

Test Steps:

1. After successfully login to Admin.
2. Add products from the add products page.

Test Case#	Test Case	Test Steps	Test Data	Expected Result
004	Verify Dashboard	1. Go to Add product page 2. Add New Product	Laptop	New Product Added Successfully.

Table 3: Test case for add products

Test Case 3:**Title:** Module Testing of View Products

Description: A user can view the products listed.

Test Steps:

1. After successfully login to Dashboard.
2. Click on the view product page.

Test Case#	Test Case	Test Steps	Test Data	Expected Result
006	Verify View Products	1. Go to Dashboard 2. Products	Products	Product can be viewed

Table 4: Test case for View Products

Test Case 4:**Title:** Module Testing of Edit Products

Description: An admin should be able to edit the products.

Test Steps:

1. After successfully login to the Admin Dashboard.
2. Go to view product and there there is edit button, Click there.

Test Case#	Test Case	Test Steps	Test Data	Expected Result
007	Verifying Edit the Products	1. Go to Dashboard 2. Go to View Products	Edit Products	Product Updated Successfully.

Table 5: Test case for Edit Products

Test Case 5:**Title:** Module Testing of Deleting Products

Description: An admin can delete the products.

Test Steps:

1. After successfully logging in to the admin dashboard
2. Products can be deleted from the view products, there is a button for deleting the product alongside the edit button.

Test Case#	Test Case	Test Steps	Test Data	Expected Result
008	Verify Delete products	1. Go to Dashboard 2. Go to view products and there is delete button. Click on that	Abc product	Product Deleted successfully

Table 6: Test case for Delete Products

Test Case 6:**Title:** Module Testing of Logout

Description: The logout operation performs. Now to page will be forwarded to the login page.

Test Steps:

1. After successfully login to the Dashboard
2. Click on Logout.

Test Case#	Test Case	Test Steps	Test Data	Expected Result
008	Verify Logout	1. Go to Dashboard 2. Click on Logout	6	Admin will exit from the admin Dashboard.

Table 7: Test case for Logout

9. Project Result

The Inventory Management System has successfully streamlined stock management, and sales tracking, significantly improving operational efficiency. Businesses can now monitor real-time stock updates, reducing the risks of overstocking or stockouts.

The role-based access control ensures that employees can efficiently process sales and view inventory, while admins maintain full control over stock, users, and reports. The integration of a sales history and customer tracking system has enhanced transparency and accountability in transactions.

By implementing smart search filters, automated inventory updates, and an intuitive user interface, the system has resolved common challenges such as inventory mismanagement, outdated stock records, and inefficient billing processes.

Overall, the project has enhanced productivity, accuracy, and security, providing businesses with a scalable and reliable inventory management solution.

10. Future Enhancements

- Integration with Multiple Payment Gateways
- Push Notifications
- User Feedback System

11. Conclusion

The successful implementation of the Inventory Management System marks a significant achievement in streamlining inventory processes. Key accomplishments include:

- A secure and user-friendly interface for real-time inventory tracking.
- A dynamic admin dashboard for seamless stock and billing management.
- Reliable and efficient updating of inventory as goods are picked up or billed.

This project showcases the effective application of web technologies such as HTML, CSS, JavaScript, PHP, and MySQL to create a functional and robust solution. Furthermore, it emphasizes the importance of a structured approach to development, incorporating iterative testing and user feedback to address practical needs. The system not only simplifies inventory management but also demonstrates scalability and adaptability for future enhancements.

12. Annexures

Our Inventory Management System has been developed with key features such as stock management, sales processing, and admin control. We have successfully achieved several key milestones, ensuring a seamless and efficient experience for users.

The system features a user-friendly interface with smart search and filtering options, allowing users to quickly locate products based on category, stock levels, and price. With real-time inventory updates, businesses can confidently track stock availability, minimizing errors and preventing overselling.

Additionally, the system enhances transparency and efficiency by providing a clear sales history, and customer transaction tracking. Employees can efficiently manage sales while admins retain full control over inventory, user roles, and reports.

Overall, these achievements have eliminated issues like stock mismanagement, outdated inventory records, and inefficient billing processes, providing a secure and reliable platform for businesses to manage their inventory with ease.

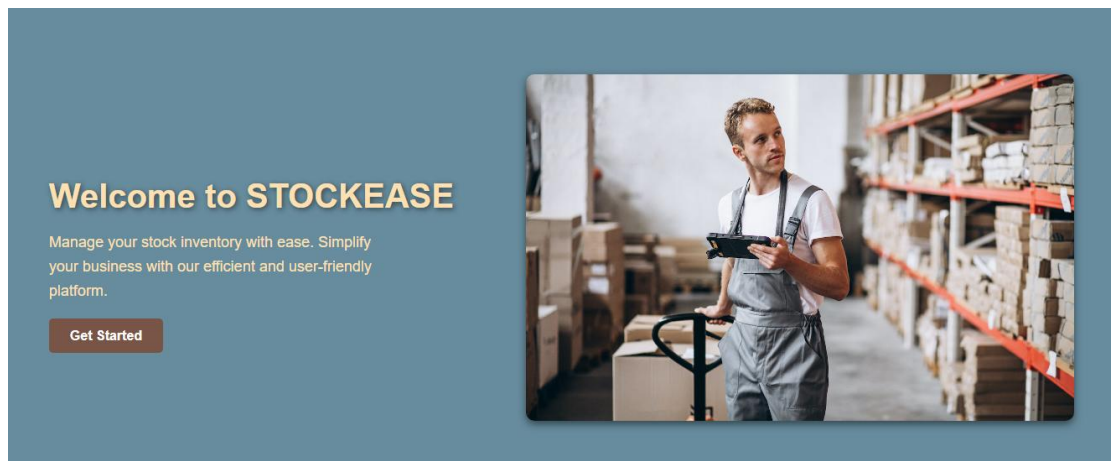


Fig 12. 1: Homepage

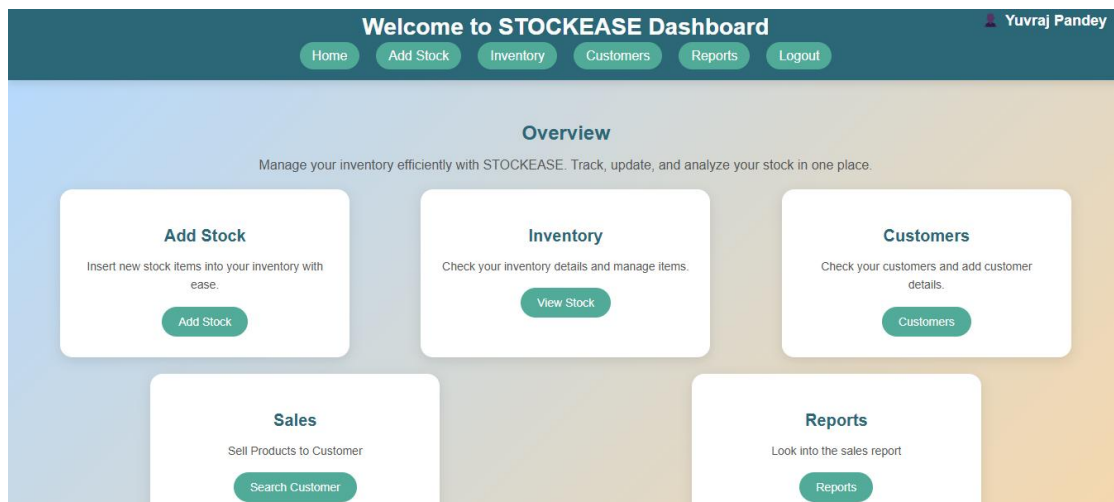


Fig 12. 2: Dashboard

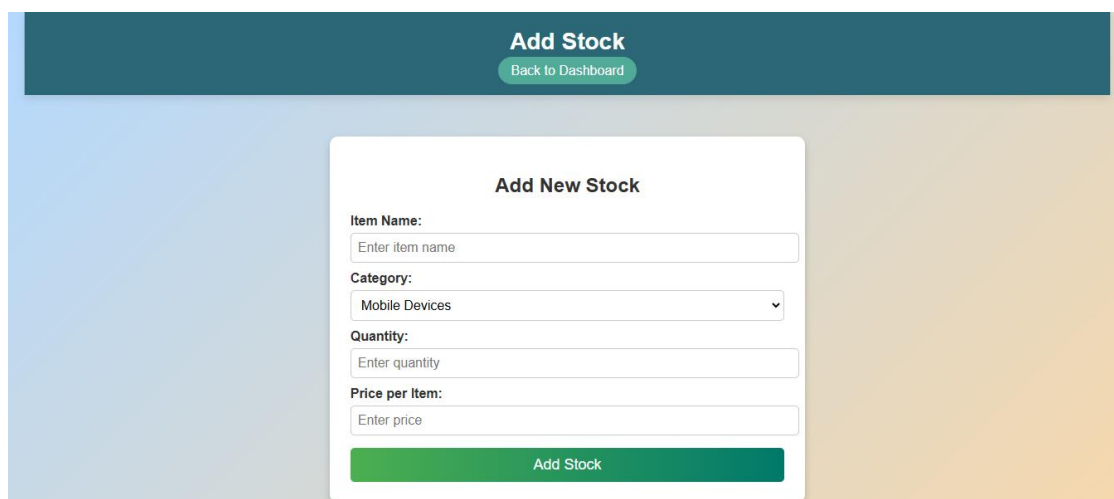


Fig 12. 3: Add Stock

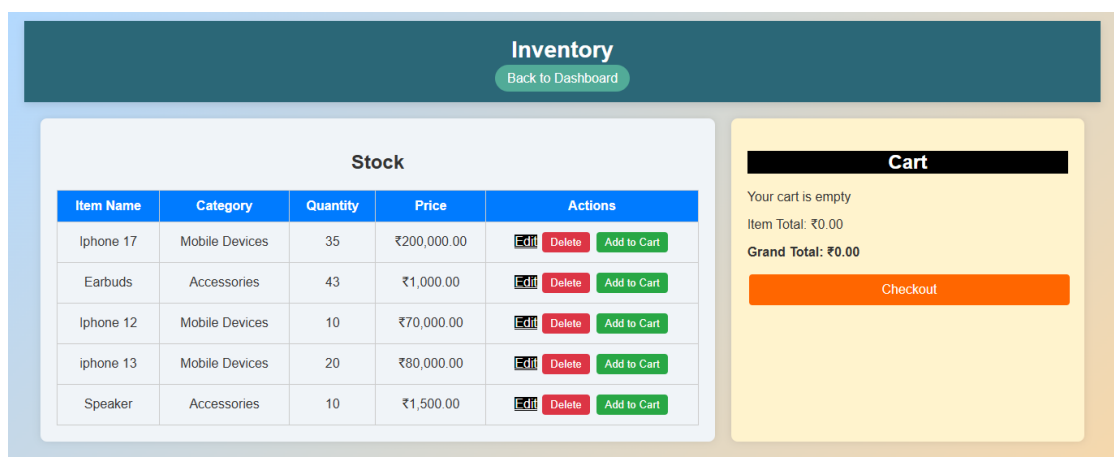


Fig 12. 4: Inventory

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